Basic Terminology

1. Array
An array is a collection of elements stored together under a single variable. In NumPy, arrays are used to efficiently store and perform mathematical operations on numerical data.

2. NumPy
NumPy (Numerical Python) is a Python library used for numerical and scientific computing. It provides powerful data structures such as arrays and functions for mathematical and statistical operations.

3. 1D Array
A one-dimensional array contains elements arranged in a single row or sequence. Example: [10, 20, 30, 40].

4. 2D Array
A two-dimensional array contains elements arranged in rows and columns. It is commonly used to represent matrices.

5. Matrix
A matrix is a rectangular arrangement of elements in rows and columns. Matrix operations include addition, multiplication and finding the determinant.

6. Indexing
Indexing is the process of accessing a particular element of an array using its position. In Python, indexing starts from 0.

7. Slicing
Slicing is used to extract a portion of an array using a range of indexes. It allows us to select specific elements without accessing them individually.

8. Reshaping
Reshaping changes the dimensions of an array without changing its data. For example, a 1D array of 10 elements can be changed into a 2 × 5 array.

9. Broadcasting
Broadcasting allows NumPy to perform arithmetic operations on arrays of different but compatible shapes. It eliminates the need for explicit loops in many operations.

10. Mean
The mean is the average value of a dataset. It is calculated by dividing the sum of all values by the total number of values.
Formula: Mean = Sum of values / Number of values

11. Median
The median is the middle value of a dataset after arranging the values in ascending or descending order. It is useful for understanding the central position of data.

12. Mode
The mode is the value that occurs most frequently in a dataset. A dataset can have one or more modes.

13. Variance
Variance measures how far the data values are spread from their mean. A higher variance indicates greater variation in the dataset.

14. Standard Deviation
Standard deviation measures the amount of variation or spread in a dataset. A small standard deviation means the values are close to the mean.

Data & Visualization

15. Dataset
A dataset is a collection of related data used for analysis or machine learning. It may contain rows representing observations and columns representing features.

16. Pandas
Pandas is a Python library used for data manipulation and analysis. It provides useful structures such as Series and DataFrame.

17. DataFrame
A DataFrame is a two-dimensional tabular data structure in Pandas consisting of rows and columns. It is commonly used to store and analyze datasets.

18. CSV
CSV (Comma-Separated Values) is a file format used to store tabular data. Each row represents a record and values are generally separated by commas.

19. Matplotlib
Matplotlib is a Python library used for creating graphs and visualizations. It can be used to create line plots, bar charts, histograms and box plots.

20. Histogram
A histogram is a graph used to represent the frequency distribution of numerical data. Data values are divided into intervals called bins.

21. Bin
A bin is an interval or range used to group values in a histogram. Each bin represents the frequency of values falling within that range.

22. Box Plot
A box plot represents the distribution and spread of a dataset using quartiles, median and extreme values. It can also help identify possible outliers.

23. Bar Chart
A bar chart represents data using rectangular bars. It is mainly used to compare values between different categories.

24. Line Plot
A line plot represents data points connected by lines. It is commonly used to show trends or changes in values over time.

Machine Learning Terminology

25. Machine Learning
Machine Learning is a technique in which computers learn patterns from data and use them to make predictions or decisions. It reduces the need for explicitly programming every rule.

26. Training Data
Training data is the portion of a dataset used to teach a machine learning model. The model learns patterns and relationships from this data.

27. Testing Data
Testing data is used to evaluate the performance of a trained machine learning model. It contains data that the model has not used during training.

28. Feature
A feature is an input variable or attribute used by a machine learning model to make predictions. For example, Age and Income can be features.

29. Label
A label is the output or target value that a machine learning model tries to predict. For example, Yes or No can be labels in a classification problem.

30. Classification
Classification is a machine learning technique used to assign data to predefined categories or classes. For example, classifying an email as spam or not spam.

31. Regression
Regression is a machine learning technique used to predict continuous numerical values. For example, predicting house prices based on different features.

32. Prediction
Prediction is the output generated by a trained machine learning model for new or unseen data. It represents the model's estimated result.

33. Accuracy
Accuracy measures how many predictions made by a classification model are correct. It is usually expressed as a percentage.
Accuracy = Correct Predictions / Total Predictions × 100

Algorithm Terminology

34. Hypothesis
A hypothesis is a possible rule or function used to describe the relationship between input data and output. In FIND-S, the goal is to find the most specific hypothesis.

35. FIND-S
FIND-S is a concept-learning algorithm that finds the most specific hypothesis

consistent with the positive training examples. It starts with the most specific hypothesis and generalizes it when necessary.

36. Perceptron
A perceptron is a basic artificial neuron used for classification. It takes inputs, applies weights and bias, and produces an output using an activation function.

37. Weight
A weight represents the importance assigned to an input in a machine learning or neural network model. During training, weights may be adjusted to improve predictions.

38. Bias
A bias is an additional value added to the weighted sum of inputs in a neural network. It helps the model adjust the output independently of the input values.

39. Naïve Bayes
Naïve Bayes is a probability-based classification algorithm based on Bayes' theorem. It assumes that the input features are conditionally independent.

40. Decision Tree
A Decision Tree is a tree-based machine learning model used for classification or prediction. It makes decisions by repeatedly applying conditions to the features.

41. ID3
ID3 (Iterative Dichotomiser 3) is a decision-tree algorithm that selects attributes using entropy and information gain. It is mainly used for classification.

42. Entropy
Entropy is a measure of uncertainty or impurity in a dataset. In ID3, the attribute that provides the best reduction in entropy is selected for splitting.

43. KNN
KNN (K-Nearest Neighbors) is a classification algorithm that classifies a new data point based on the classes of its nearest neighboring points.

44. K
In KNN, K represents the number of nearest neighbors considered when classifying a new data point. The appropriate value of K can affect the accuracy of the model.

45. K-Means
K-Means is an unsupervised machine learning algorithm used to divide data into a specified number of groups called clusters. It assigns data points based on their distance from cluster centers.

46. Cluster
A cluster is a group of similar data points formed during clustering. Data points within the same cluster are more similar to each other than to points in other clusters.

47. Centroid
A centroid is the center point of a cluster in the K-Means algorithm. Data points are assigned to the cluster whose centroid is closest.

48. Neural Network
A Neural Network is a machine learning model made up of interconnected artificial neurons. It consists of layers such as input, hidden and output layers.

49. Backpropagation
Backpropagation is a training algorithm used in neural networks to reduce errors. It calculates the error and adjusts the network's weights accordingly.

50. Activation Function
An activation function determines the output of a neuron based on its input. It introduces non-linearity and helps neural networks learn complex patterns.

Most Important for Practical Exam

NumPy → Array → Indexing → Slicing → Broadcasting → Pandas → DataFrame → Dataset → Mean → Median → Mode → Variance → Standard Deviation → Histogram → Box Plot → Regression → Classification → Training Data → Testing Data → Feature → Label → Perceptron → Naïve Bayes → Entropy → ID3 → KNN → K-Means → Cluster → Centroid → Neural Network → Backpropagation.